📘 SQL CLASS NOTE – MODULE 3: `WHERE`

1. Introduction

The **`WHERE` clause** is used to **filter records**.
It allows you to set conditions so that SQL only returns rows that meet your criteria.

- Without `WHERE` , you get **everything**.
- With `WHERE` , you get **only what you need**.

👉 Think of `WHERE` as a **filter**:

- "Show me only Grade 10 students."
- "Show me all courses taught by Dr. Wilson."
- "Show me enrollments after January 2025."

2. Syntax

```
SELECT column1, column2, ...
FROM table_name
WHERE condition;
```

- `condition` → a logical test (true/false).
- If the row meets the condition, it appears in the results.

3. Operators You Can Use in `WHERE`

Operator	Description	Example
`=`	Equals	`WHERE Gender = 'Male'`
`!=` or `<>`	Not equal	`WHERE Age <> 16`

Operator	Description	Example
`>`	Greater than	`WHERE Age > 16`
`<`	Less than	`WHERE Age < 18`
`>=`	Greater or equal	`WHERE Age >= 17`
`<=`	Less or equal	`WHERE Age <= 15`
`AND`	Both conditions true	`WHERE Age > 15 AND Gender = 'Female'`
`OR`	At least one true	`WHERE GradeLevel = 'Grade 9' OR GradeLevel = 'Grade 10'`
`BETWEEN`	Range	`WHERE Age BETWEEN 15 AND 17`
`IN`	Match multiple values	`WHERE GradeLevel IN ('Grade 10','Grade 11')`
`LIKE`	Pattern matching	`WHERE FirstName LIKE 'J%'`
`IS NULL`	Empty values	`WHERE LastName IS NULL`
`IS NOT NULL`	Not empty	`WHERE LastName IS NOT NULL`

4. Practical Examples with `SchoolDB`

Example 1: Select all female students

```sql
SELECT FirstName, LastName, Gender
FROM Students
WHERE Gender = 'Female';
```

Example 2: Select students younger than 16

```sql
SELECT FirstName, Age
FROM Students
WHERE Age < 16;
```

Example 3: Select students in Grade 10 or Grade 11

```sql
SELECT FirstName, LastName, GradeLevel
FROM Students
WHERE GradeLevel IN ('Grade 10', 'Grade 11');
```

Example 4: Select enrollments in 2025 only

```sql
SELECT StudentID, CourseID, EnrollmentDate
FROM Enrollments
WHERE YEAR(EnrollmentDate) = 2025;
```

Example 5: Pattern matching with `LIKE`

```sql
SELECT FirstName, LastName
FROM Students
WHERE FirstName LIKE 'J%';
```

- Finds names starting with **J** (e.g., John, James).

Example 6: Using multiple conditions

```sql
SELECT FirstName, Age, Gender
FROM Students
WHERE Age > 15 AND Gender = 'Male';
```

- Male students older than 15.

Example 7: Range with `BETWEEN`

```sql
SELECT FirstName, Age
FROM Students
WHERE Age BETWEEN 15 AND 17;
```

- Students aged 15, 16, or 17.

5. Why is `WHERE` Important?

- Helps retrieve **only relevant data**.
- Saves **time** and **resources** by not fetching unnecessary rows.
- Essential for **reporting** and **analytics**.

6. Real-Life Use Cases

- Schools: Find students in a particular grade.
- Banking: Find all transactions above ₦50,000.
- Hospitals: Retrieve patients admitted after a certain date.
- E-commerce: Show products below a certain price.

7. Classwork (Practice) ✅

👉 Try these queries on **SchoolDB**:

1. Get all students who are **16 years old**.
2. Show all students in **Grade 9**.
3. Find all students whose names start with **"M"**.
4. List all enrollments that happened **before 2025-01-15**.
5. Show students who are **female** and **older than 16**.
6. Find all courses taught by **Dr. Wilson**.
7. Get all students whose `GradeLevel` is **Grade 11** or **Grade 12**.

8. Challenge Question 💡

👉 Write a query to display the **FullName**, `Age`, and `GradeLevel` of all students:

- Who are **between 15 and 17 years old**
- **AND** are in either **Grade 10 or Grade 11**

✅ That’s the **expanded Module 3:** `WHERE` .

Perfect 👍 Let’s solve the **Module 3 Classwork + Challenge** step by step using `SchoolDB` .

✅ Solutions to Module 3: `WHERE`

1. Get all students who are **16 years old**.

```sql
SELECT FirstName, LastName, Age
FROM Students
WHERE Age = 16;
```

2. Show all students in **Grade 9**.

```sql
SELECT FirstName, LastName, GradeLevel
FROM Students
WHERE GradeLevel = 'Grade 9';
```

3. Find all students whose names start with “M”.

```sql
SELECT FirstName, LastName
FROM Students
WHERE FirstName LIKE 'M%';
```

📌 `%` means “any characters after M.”

4. List all enrollments that happened **before 2025-01-15**.

```
SELECT StudentID, CourseID, EnrollmentDate
FROM Enrollments
WHERE EnrollmentDate < '2025-01-15';
```

5. Show students who are **female** and **older than 16**.

```
SELECT FirstName, LastName, Age, Gender
FROM Students
WHERE Gender = 'Female' AND Age > 16;
```

6. Find all courses taught by **Dr. Wilson**.

```
SELECT CourseName, Teacher
FROM Courses
WHERE Teacher = 'Dr. Wilson';
```

7. Get all students whose `GradeLevel` is **Grade 11** or **Grade 12**.

```
SELECT FirstName, LastName, GradeLevel
FROM Students
WHERE GradeLevel IN ('Grade 11', 'Grade 12');
```

💡 Challenge Question

👉 Display the **FullName**, `Age` , and `GradeLevel` of all students:

- Who are **between 15 and 17 years old**
- **AND** are in either **Grade 10 or Grade 11**

```sql
SELECT CONCAT(FirstName, ' ', LastName) AS FullName,
       Age,
       GradeLevel
FROM Students
WHERE Age BETWEEN 15 AND 17
  AND GradeLevel IN ('Grade 10', 'Grade 11');
```

✅ This combines multiple filters to give a **targeted result**.

✨ Now we've wrapped up `WHERE` with practice and challenge solved.

👉 Should I move on to **Module 4:** `ORDER BY` (sorting results) in the same detailed, practical style?